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Micro-electro-mechanical systems:

It Is Good To SEPARATE FAST- And SLOW- MOVING TECHNOLOGIES

A vast expanse of applications is driving innovation in micro-electro-mechanical systems (MEMS), feels Uday Prabhu, general manager, electronics product engineering and product management, Robert Bosch Engineering and Business Solutions Pvt Ltd. Let us find out how as he talks to Dilin Anand and Priya Ravindran of EFY about the growth and scope of MEMS technology

Q. How has MEMS technology grown in the last five years?

A. From a research and development perspective, MEMS has seen an explosion of sorts and quite a few diversifications. It started proliferating in the automotive segment, with regards to electronic stability programme.

The technology then exploded in the consumer space, with increasing demand for smartphones, gaming devices and wearables, so much so that companies currently manufacture about five million MEMS sensors each day.

Q. What are the primary enablers for the improvement of sensors?

A. MEMS is about how best we can integrate mechanical structures into printing on electronic substrates. People are now looking at the fundamental physics of sensing. The number of axes that we use determines the kind of resolution we can achieve with respect to gathering information about the moment, and we see an expansion in the sensing dimensions.

New sensing elements like pressure, humidity and other environmental parameters are coming

out. Any physical element that can be leveraged for sensing is now being transformed into MEMS. Also, upgrades in the process of production have resulted in smaller form factors.

Q. Please tell us more about these upgrades?

A. There is compression happening on a large scale. An increase in the number of transistors that can go into a chip has resulted in reduced size with higher functionality. Earlier, MEMS elements functioned exclusively as sensing elements.